

COMS W4156 Advanced Software Engineering (ASE)

October 29, 2024

On this day ...



Scientific Discovery

1998 Space Shuttle Discovery blasts off on STS-95 with 77-year old [John Glenn](#) on board, making him the oldest person to go into space.



Astronaut and Politician
[John Glenn](#)

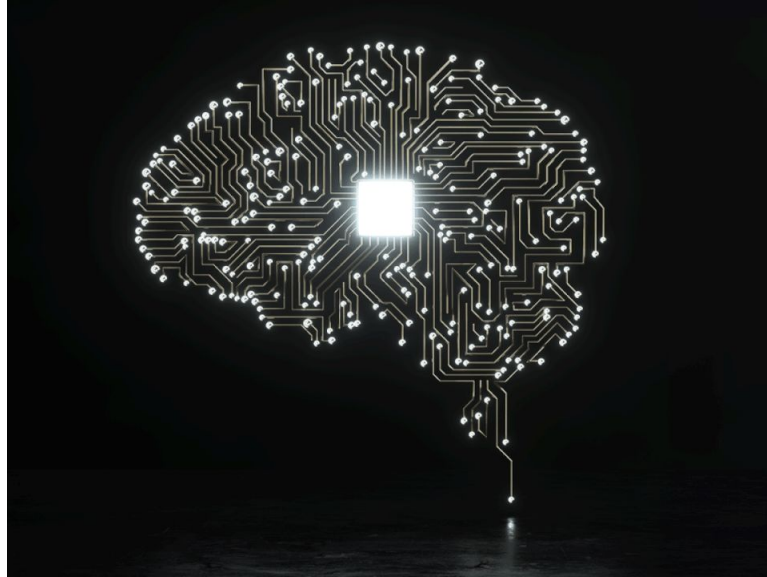
Agenda

1. More AI presentations: Amy Pu and Sree Rammohan
2. Second Iteration starts now
3. Next



Amy Pu and Sree Rammohan AI Presentations

[slides](#)



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The text "SECOND ROUND" is rendered in a pixelated, blocky font. The word "SECOND" is in blue with a black outline, and the word "ROUND" is in orange with a black outline. The letters have a slightly 3D effect with a darker shade on the bottom.

Second Iteration

Due November 25 - 4 weeks to build version 2 and sample client

Version 2 of your service should be your final version (except for bug fixes)



Remember the Team Project Constraints



Your team's service must be a service, not a library and not an app

The first iteration focused entirely on the service, the second iteration adds a client that uses the service's API

The client may be either a service or an app (or even a library, if you can find some way to demo)

Your team project must be developed in Java or C++, Python is not allowed except for AI components (if any)

The client depends on the service but the service must not depend on the client

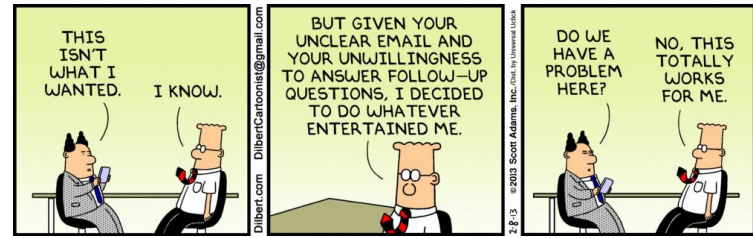
The client must target an acceptable user group that is not university students, staff or faculty (except for individuals who happen to also be members of an acceptable user group)

What Is An Acceptable User Group?



- Elderly (defined for this purpose as age 75 or over)
- Disabled or impaired (e.g., vision, hearing, speech, motor, cognitive)
- Healthcare workers (doctors, medical technicians, nurses, aides, etc.)
- First responders (police, fire, EMT)
- Indigenous peoples ("distinct social and cultural groups that share collective ancestral ties to the lands and natural resources where they live, occupy or from which they have been displaced")
- Homeless people ("lacks a fixed, regular, and adequate nighttime residence")
- Maybe others with societal benefits - ask me *in advance*

Second Iteration Overview



Final version of your service's API. If the API is not the same as proposed in your revised proposal, your README should explain what is different and why

Demoable client that uses your service's API and targets one (or more) of the designated target audiences

Service supports multiple instances of the same client running simultaneously (show this in your demo)

Client code should reside in the same repository as your service or in another repository linked from the README of your service repository. Make sure the instructor and your Mentor are collaborators on both repos

Second Iteration Testing



Update unit test suite (with mocking) to run under CI

Update API test suite to run under CI and exercise your service's complete final API, including support for multiple clients and persistent data

Add internal integration testing to run under CI to cover direct and indirect dependencies within your service implementation

Add external integration testing to run under CI to cover dependencies on third-party code (if any) and external resources (files, databases, devices if applicable) within your service implementation

Collectively, your automated tests under CI should achieve $\geq 85\%$ branch coverage

Add end-to-end testing where your client exercises your service's API, manual is ok

Second Iteration Documentation



Document what your client does and how it helps its target audience

Document your API entry points, including ordering rules and all possible errors and exceptions - either prose or a special format like OpenAPI/Swagger is ok

Document both your own client's use of your service's API and what a third party would need to do (and not do) to develop their own client that uses your service's API

Document your tests, particularly integration tests - which specific interfaces are they testing? Where did the inputs come from? How do you know the outputs and side-effects are correct? Consider all inputs and outputs, such as shared data, not just parameters and return values

Second Iteration Odds and Ends

Make sure everything that matters is in the main branch of your repository(ies), your Mentor will not check other branches

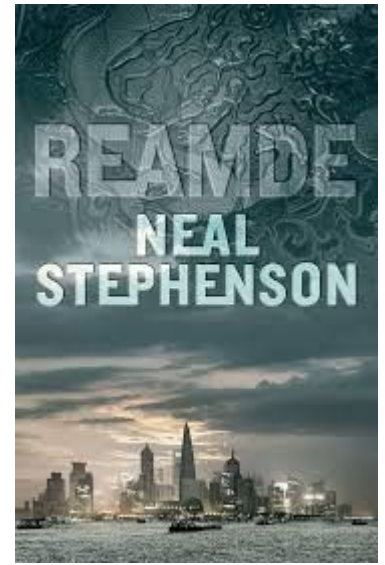
Your README(s) should include instructions about how to build and run both your service and your client, and instructions about how to perform any manual testing

Your main README should include a link to your project management task board

Run style checker and static analysis bug finder under CI on your entire service codebase (not required for client) and fix as many flaws as possible

Include test runner, coverage, static checker, etc. reports in your service repository

Explain any third-party code in your repository(ies): what is it, where did you get it, what are you using it for?



Second Iteration Demo

Due December 6

Demo for your mentor sometime during the approx. two weeks following the second iteration deadline (note Thanksgiving break is in the middle)

You can continue working on your project between the second iteration deadline and the second iteration demo (and Demo Day later), but focus on fixing bugs and presenting an impressive demo



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Thursday

AI-based Team Project Demo

CRC Design

What You Need To Do For Your Second Iteration (again)



Ask Me Anything

